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Multi-axial unsplit frequency-shifted perfectly matched layer for displacement-based anisotropic wave simulation in infinite domain

Xie Zhinan[1, 2†], Zheng Yonglu[1, 2‡], Paul Cristini[3§] and Zhang Xubin[1, 2*]

1. *Key Laboratory of Earthquake Engineering and Engineering Vibration, Institute of Engineering Mechanics, China Earthquake Administration, Harbin 150080, China*
2. *Key Laboratory of Earthquake Disaster Mitigation, Ministry of Emergency Management, Harbin 150080, China*
3. *Aix Marseille University, CNRS, Centrale Marseille, LMA, Marseille F-13353, France*

Abstract: Multi-axial perfectly matched layer (M-PML), known to have lost the perfect-matching property owing to multi-axial coordinate stretching, has been numerically validated to be long-time stable and it is thus used extensively in linear anisotropic wave simulation and in isotropic cases where the PML becomes unstable. We are concerned with the construction of the M-PML for anisotropic wave simulation based on a second order wave equation implemented with the displacement-based numerical method. We address the benefit of the incorrect chain rule, which is implicitly adopted in the previous derivation of the M-PML. We show that using the frequency-shifted stretching function improves the absorbing efficiency of the M-PML for near-grazing incident waves. Then, through multi-axial complex-coordinate stretching the second order anisotropic wave equation in a weak form, we derive a time-domain multi-axial unsplit frequency-shifted PML (M-UFSPML) using the frequency-shifted stretching function and the incorrect chain rule. A new approach is provided to reduce the number of memory variables needed for computing convolution terms in the M-UFSPML. The obtained M-UFSPML is well suited for implementation with a finite element or the spectral element method. By providing several typical examples, we numerically verify the accuracy and long-time stability of the implementation of our M-UFSPML by utilizing the Legendre spectral element method.

Keywords: computational seismology; seismic anisotropy; wave propagation; elastodynamics

1 Introduction

Linear elastic anisotropy is common to wave propagation in the Earth′s interior (Babuška and Cara, 1991; Savage, 1999; Deuss *et al*., 2010; Almqvist and Mainprice, 2017) and its ice cover (Diez and Eisen, 2015; Sayers, 2018). Such anisotropy can be caused by factors such as a preferred orientation of crystals, aligned inclusions, or regular sequences of thin layers (Cerveny, 2005). In anisotropic media, seismic wave speeds depend on the local directions of wave propagation and wave polarization. In recent years, wave simulation integrated as part of adjoint seismic full-waveform tomography has increasingly been used in investigating the regional-scale anisotropic structure of the Earth (Zhu *et al*., 2017).

In wave simulation, taking anisotropy into account makes it difficult to establish stable techniques for infinite domain truncation (Komatitsch *et al*., 2000). It is now well-known that the Perfectly Matched Layer (PML Bérenger, 1994) is intrinsically unstable for certain types of anisotropic media (Bécache *et al*. 2003). Such instability is irrelevant to the type of complex coordinate stretching functions used in deriving the PML and the PML formulation, such as a split, unsplit or auxiliary differential PML formulation derived from either the velocity-stress-based wave equation in the first-order form or the displacement-based wave equation in the second-order form (Komatitsch and Martin, 2007; Ping *et al*., 2014).

In the first-order case, Meza-Fajardo and Papageorgiou (2008) proposed an approach to stabilize the split PML and produce what is referred to as a multi-axial PML (M-PML) in the anisotropic-wave simulation. Zeng *et al*. (2011) showed that the M-PML can stabilize a wave simulation for a horizontally layered

Correspondence to: Xie Zhinan, Institute of Engineering Mechanics, China Earthquake Administration, 29 Xuefu Road, Harbin 150080, China
Tel: +86-15114557253
E-mail: wla_2012@163.com
†Professor; ‡Master; §Research Associate; *Associate Professor

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isotropic solid medium that has a high Poisson ratio. In the M-PML, aside from damping in the direction that is customary for interfaces situated between the M-PML and the truncated domain, additional nonzero damping in tangential directions is introduced. Although tangential damping is usually set to be much smaller than that of normal damping, an important drawback is that the M-PML is not perfectly matched (Dmitriev and Lisitsa, 2011). Therefore, the M-PML can be interpreted as a compromise between the PML and absorbing layers, with the same level of tangential and normal damping (Cerjan *et al*., 1985; Sochacki *et al*., 1987; Semblat *et al*., 1987; Halpern *et al*., 2011). To achieve the same level of accuracy, M-PML is often more computationally efficient than is the case with standard absorbing layers, but less efficient than PML. Gao and Huang (2017) developed a numerical algorithm to determine the optimal level of tangential damping.

First-order M-PML cannot be directly used in the second-order wave equation simulation with displacement-based numerical simulation methods, such as the finite element (FE) method (e.g., Bao *et al*., 1998), the spectral element (SE) method (e.g., Komatitsch and Tromp, 1999), or some finite difference methods (e.g., Moczo *et al*., 2002). Festa and Vilotte (2005) showed how to couple the first-order split PML, which was implemented by using the staggered finite difference method. The first-order split PML was implanted into the second-order wave equation, implemented with the FE/SE method and the Newmark-beta time-stepping scheme, by using the equivalence between the Newmark time-stepping scheme and the midpoint difference rule. Following Festa and Vilotte′s observation, Martin *et al*. (2008) extended the first-order convolutional frequency-shifted PML to the multi-axial case and then coupled it with the second-order wave equation simulation, employing the SE method and the Newmark-beta time-stepping scheme. Such a coupling remains an open problem in the sense that other time-stepping schemes are used (Liu and Zhang, 2019).

Following the idea of Meza-Fajardo and Papageorgiou (2008), several second-order M-PMLs have been derived by using multi-axial coordinate stretching for the second-order wave equation. Li and Bou (2010) developed a second-order convolutional frequency-shifted M-PML in the frequency domain. In principle, Li and Bou's formulation is not fit for layered heterogenous infinite-domain truncation because they treated the gradient of all convolutional terms in stress as additional body forces. Ping *et al*. (2014, 2016) extended the second-order split PML proposed by Komatitsch and Tromp (2003) for the multi-axial case. The formulation of Ping *et al*. (2014, 2016) inherited the shortcomings of the PML of Komatitsch and Tromp (2003), arising from the introduced hybrid second-order and third-order ordinary differential equations. The number of split displacement variables is at least five times that of the original variables (Ma and Liu, 2006). Moreover, Ping *et al*. (2014, 2016) obtained an inconsistent result that, unlike the obtained M-PML with a damping parameter defined in the high-order polynomial profile, the M-PML with a damping parameter defined in the commonly used second-order polynomial profile loses its stability in a long-term simulation. Fathi *et al*. (2015) showed without theoretical justification that although the chain rule prevalently used in the M-PML derivation is incorrect, the resulting M-PML performed better than the M-PML derived by utilizing the corrected chain rule.

In this paper, we develop a second-order multi-axial unsplit frequency-shifted PML (M-UFSPML). We first show the importance of introducing the non-rigorous chain rule to stabilize M-UFSPML. Additionally, the introduction of frequency-shifted coordinate stretching also can improve the absorbing efficiency of the M-PML. We then derive the M-UFSPML by multi-axial complex-coordinate stretching the second-order wave equation in its weak form—rather than using the strong wave equation—to avoid possible improper matching between the obtained PML wave equation and boundary or interface conditions. Improper matching is an important mechanism because it leads to an unstable PML (Duru and Kreiss, 2014). We introduce a new way to reduce the number of memory variables needed for dealing with the computation of convolution terms. The obtained M-UFSPML is in a weak form and thus is ready for implementation using the FE/SE method. We numerically verify the accuracy and long-time stability of the M-UFSPML through theoretical and practical applications in wave simulation with the high-order Legendre spectral element method and the explicit time-stepping scheme.

2 Notes on the choice of multi-axial coordinate stretching

Dmitriev and Lisitsa (2011) showed that the loss of the perfectly matched property of M-the PML is due to multi-axial coordinate stretching. Based on their work, we investigate the effects of multi-axial coordinate stretching, frequency-shifted coordinate stretching, and the implicitly adopted incorrect chain rule on the absorbing efficiency of the M-PML, which leads to a useful remark regarding our late M-PML derivation.

For the simplest wave simulation in $(x,y)\in R$, we specify the computation domain as $\{x<0, y\in R\}$. Within this domain, the wave equation is:

$$\begin{bmatrix} \mathrm{i}\omega\hat{u}_x \\ \mathrm{i}\omega\hat{u}_y \\ \mathrm{i}\omega\hat{p} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1/\rho \\ 0 & 0 & 0 \\ -\rho c^2 & 0 & 0 \end{bmatrix} \begin{bmatrix} \partial_x u_x \\ \partial_x u_y \\ \partial_x p \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1/\rho \\ 0 & -\rho c^2 & 0 \end{bmatrix} \begin{bmatrix} \partial_y \hat{u}_x \\ \partial_y \hat{u}_y \\ \partial_y \hat{p} \end{bmatrix} \tag{1}$$

where ρ is the density; c is acoustic wave speed; u_x and u_y are the displacements in the x and y directions, respectively; and p is pressure. A caret above a variable denotes a Fourier transformation. And $\mathrm{i}=\sqrt{-1}$ and ω are the circular frequency. Following Meza-Fajardo and Papageorgiou (2008), we introduce complex coordinates in $\{x\geq 0, y\in R\}$ as:

$$\begin{aligned}\tilde{x}(x,y)&=x+\int_0^x\left[d(x)/\left[\alpha(x)+\mathrm{i}\omega\right]\right]\mathrm{d}x',\\ \tilde{y}(x,y)&=\left[1+p^{y/x}\,d(x)/\left[\alpha(x)+\mathrm{i}\omega\right]\right]y\end{aligned}\tag{2}$$

where $d(x)$ is the damping profile in the x direction, and the constant $p^{y/x}$ refers to the level of additionally introduced damping in the y direction. Also, $\alpha(x)$ serves to enhance the attenuation of evanescent and near-grazing waves and was set to zero by Meza-Fajardo and Papageorgiou (2008). With the incorrect chain rules provided by Meza-Fajardo and Papageorgiou (2008), we obtain:

$$\begin{aligned}&\mathrm{d}\tilde{x}=s_x\mathrm{d}x,\ \mathrm{d}\tilde{y}=s_y\mathrm{d}y,\ \mathrm{d}\tilde{x}\mathrm{d}\tilde{y}=s_xs_y\mathrm{d}x\mathrm{d}y,\\ &\partial_{\tilde{x}}=(1/s_x)\partial_x,\ \partial_{\tilde{y}}=(1/s_y)\partial_y\end{aligned}\tag{3}$$

where $s_x=1+d(x)/\left[\alpha(x)+\mathrm{i}\omega\right]$ and $s_y=1+p^{y/x}\,d(x)/\left[\alpha(x)+\mathrm{i}\omega\right]$, denoting the coordinate stretching factor. We obtain the frequency-shifted M-PML by first mapping Eq. (1) to complex coordinates and then transforming to it real coordinates using Eq. (3):

$$\begin{bmatrix}\mathrm{i}\omega\hat{u}_x\\ \mathrm{i}\omega\hat{u}_y\\ \mathrm{i}\omega\hat{p}\end{bmatrix}=\begin{bmatrix}0&0&-1/(\rho s_x)\\ 0&0&0\\ -\rho c^2/s_x&0&0\end{bmatrix}\begin{bmatrix}\partial_x u_x\\ \partial_x u_y\\ \partial_x p\end{bmatrix}+\begin{bmatrix}0&0&0\\ 0&0&-1/(\rho s_y)\\ 0&-\rho c^2/s_y&0\end{bmatrix}\begin{bmatrix}\partial_y\hat{u}_x\\ \partial_y\hat{u}_y\\ \partial_y\hat{p}\end{bmatrix}\tag{4}$$

However, because s_y depends on the variable x, in the corrected sense, the chain rule should be written as:

$$\begin{aligned}&\mathrm{d}\tilde{x}=s_x\mathrm{d}x,\ \mathrm{d}\tilde{y}=\overline{s}_x\mathrm{d}x+s_y\mathrm{d}y,\ \mathrm{d}\tilde{x}\mathrm{d}\tilde{y}=s_xs_y\mathrm{d}x\mathrm{d}y,\\ &\partial_{\tilde{x}}=\frac{1}{s_x}\partial_x,\ \partial_{\tilde{y}}=-\frac{\overline{s}_x}{s_xs_y}\partial_x+\frac{1}{s_y}\partial_y\end{aligned}\tag{5}$$

where $\overline{s}_x=y\partial_x s_y(x,y')$. Following the same process, the M-PML derived with Eq. (5) can be written as:

$$\begin{bmatrix}\mathrm{i}\omega\hat{u}_x\\ \mathrm{i}\omega\hat{u}_y\\ \mathrm{i}\omega\hat{p}\end{bmatrix}=\begin{bmatrix}0&0&-1/(\rho s_x)\\ 0&0&\overline{s}_x/(\rho s_x s_y)\\ -\rho c^2(s_y-\overline{s}_x)/s_xs_y&0&0\end{bmatrix}\begin{bmatrix}\partial_x\hat{u}_x\\ \partial_x\hat{u}_y\\ \partial_x\hat{p}\end{bmatrix}+\begin{bmatrix}0&0&0\\ 0&0&-1/(\rho s_y)\\ 0&-\rho c^2/s_y&0\end{bmatrix}\begin{bmatrix}\partial_y u_x\\ \partial_y u_y\\ \partial_y p\end{bmatrix}\tag{6}$$

Following the plane-wave analysis presented by Dmitriev and Lisitsa (2011), we obtain the same reflection coefficient along the interface of the M-PML, as presented in Eq. (4):

$$R(\theta,s_y)=\left|\frac{\cos\theta-\sqrt{1-\sin^2\theta/s_y^2}}{\cos\theta+\sqrt{1-\sin^2\theta/s_y^2}}\right|\tag{7}$$

where $|\,|$ denotes the absolute norm. The reflection coefficient along the interface of the M-PML presented in Eq. (6) is:

$$R(\theta,s_y,\overline{s}_x)=\left|\frac{\cos\theta-\dfrac{\sin\theta\overline{s}_x+\sqrt{(\sin\theta)^2(\overline{s}_x^2+4s_y\overline{s}_x-4s_y^2)+4s_y^3(s_y-\overline{s}_x)}}{2s_y(s_y-\overline{s}_x)}}{\cos\theta+\dfrac{\sin\theta\overline{s}_x+\sqrt{(\sin\theta)^2(\overline{s}_x^2+4s_y\overline{s}_x-4s_y^2)+4s_y^3(s_y-\overline{s}_x)}}{2s_y(s_y-\overline{s}_x)}}\right|\tag{8}$$

Taking the commonly used profile $\alpha(x)=0$, $d(x)=(3c/2L)(|x|/L)^2$ with $c=1500$ m/s and $L=10\times c/f_0$, Fig. 1 shows the distributions of $R(\theta,s_y)$ and $R(\theta,s_y)-R(\theta,s_y,\overline{s}_x)$ evaluated at $x=L/2$ and $y=L/2$. The results show the clear loss of the perfect-matching property in the M-PML, particularly for a near-grazing incident wave. The M-PML derived using the incorrect chain rule performs almost the same as that derived with the corrected chain rule but performs better in the region around $\omega=0$. Moreover, we present in Fig. 2 the distribution of $R(\theta,s_y)$ with the same set of parameters considering commonly used nonzero $\alpha(x)=3\pi f_0(1-|x|/L)$. The figure clearly shows that nonzero $\alpha(x)$ serves to improve the absorbing efficiency of the near-grazing incident wave. Thus, in the later construction of the second-order M-PML, we reuse the non-rigorous chain rule but take into account the nonzero $\alpha(x)$.

3 The second-order M-UFSPML

By multi-axial complex stretching the second-order anisotropic wave equation in its weak form, we derive the

second-order M-UFSPML with the incorrect chain rule and frequency-shifted coordinate stretching function. Moreover, we introduce a new way to approximately halve the memory cost in handling the computation of convolution terms via memory variables. At the end of the section, we summarize M-PML formulations, and the number of additional memory variables needed for their implementation.

3.1 Second-order anisotropic wave equation

Following the matrix notation used by Carcione (2015), the second-order anisotropic wave equation can be written as:

$$\rho\frac{\partial^2\boldsymbol{u}}{\partial t^2}=\nabla\cdot\boldsymbol{\sigma}+\boldsymbol{f} \tag{9}$$

where $\boldsymbol{u}=[u_x \quad u_y \quad u_z]^{\mathrm{T}}=[u_1 \quad u_2 \quad u_3]^{\mathrm{T}}$ and $\boldsymbol{f}=[f_1 \quad f_2 \quad f_3]^{\mathrm{T}}$ are respectively vectors of the displacement and body force and

$$\begin{aligned}\boldsymbol{\sigma}&=[\sigma_{xx} \quad \sigma_{yy} \quad \sigma_{zz} \quad \sigma_{yz} \quad \sigma_{xz} \quad \sigma_{xy}]^{\mathrm{T}}\\&=[\sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \sigma_5 \quad \sigma_6]^{\mathrm{T}},\end{aligned}$$

$$\begin{aligned}\boldsymbol{\sigma}&=\boldsymbol{C}\cdot\boldsymbol{e}=\boldsymbol{C}\cdot\boldsymbol{\nabla}^{\mathrm{T}}\cdot\boldsymbol{u}\\\boldsymbol{\nabla}&=\begin{pmatrix}\partial_1 & 0 & 0 & 0 & \partial_3 & \partial_2\\0 & \partial_2 & 0 & \partial_3 & 0 & \partial_1\\0 & 0 & \partial_3 & \partial_2 & \partial_1 & 0\end{pmatrix}\end{aligned} \tag{10}$$

Here, $\boldsymbol{e}=[e_{xx} \quad e_{yy} \quad e_{zz} \quad e_{yz} \quad e_{xz} \quad e_{xy}]^{\mathrm{T}}$

$=[e_1 \quad e_2 \quad e_3 \quad e_4 \quad e_5 \quad e_6]^{\mathrm{T}}$

and $\boldsymbol{C}$ is the elasticity matrix. Let us restrict ourselves to the anisotropic medium with orthorhombic symmetry, which plays the most important role in seismology and seismic exploration. It covers the isotropic medium and the anisotropic medium of vertical transverse isotropy (VTI) symmetry, horizontal transverse isotropy (HTI) symmetry and hexagonal symmetry. The elasticity matrix of an orthorhombic medium reads:

$$\boldsymbol{C}=\begin{pmatrix}c_{11} & c_{12} & c_{13} & 0 & 0 & 0\\c_{12} & c_{22} & c_{23} & 0 & 0 & 0\\c_{13} & c_{23} & c_{33} & 0 & 0 & 0\\0 & 0 & 0 & c_{44} & 0 & 0\\0 & 0 & 0 & 0 & c_{55} & 0\\0 & 0 & 0 & 0 & 0 & c_{66}\end{pmatrix} \tag{11}$$

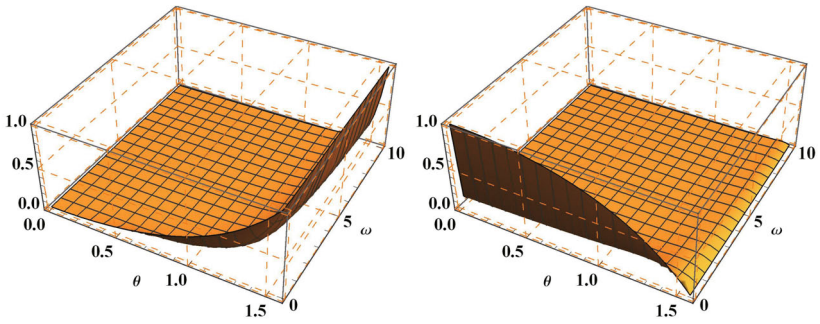


Fig. 1 Distributions of $R(\theta,s_y)$ (left) and $R(\theta,s_y)-R(\theta,s_y,\bar{s}_x)$ (right) computed with $\alpha(x)=0$, $d(x)=(3c/2L)(|x|/L)^2$ ($c=1500$ m/s), $L=10\times c/f_0$, $p^{y/x}=0.1$, $x=L/2$ and $y=L/2$

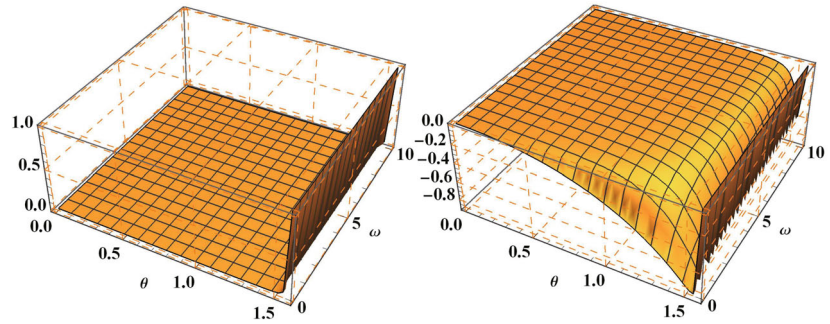


Fig. 2 Distributions of $R(\theta,s_y)$ (left) computed with the same set of parameters as in Fig. 1, except for $\alpha(x)=3\pi f_0(1-|x|/L)$ and the distribution of $R(\theta,s_y)\big|_{\alpha\neq 0}-R(\theta,s_y)\big|_{\alpha=0}$ (right)

In the two-dimensional case (having x–y Cartesian coordinates), the in-plane wave equation can be obtained by simply omitting the variables and the spatial variation related to the z coordinate. We then have $\boldsymbol{u}=\left[u_1 \quad u_2\right]^{\mathrm{T}}$, $\boldsymbol{\sigma}=\left[\sigma_1 \quad \sigma_2 \quad \sigma_6\right]^{\mathrm{T}}$, $\boldsymbol{e}=\left[e_1 \quad e_2 \quad e_6\right]^{\mathrm{T}}$,

$$\boldsymbol{\nabla}=\begin{pmatrix}\partial_1 & 0 & \partial_2\\ 0 & \partial_2 & \partial_1\end{pmatrix},\ \text{and}\ \boldsymbol{C}=\begin{pmatrix}c_{11} & c_{12} & 0\\ c_{12} & c_{22} & 0\\ 0 & 0 & c_{66}\end{pmatrix} \tag{12}$$

3.2 Derivation of the second-order M-UFSPML

The PML is derived classically through the complex coordinate stretching of the wave equation written in strong form. Recently, the weak-form approach for the PML derivation has been introduced. In the weak-form approach, the PML is derived through the complex coordinate stretching of the wave equation written in the weak form (Bindel and Govindjee, 2005; Matuszyk and Demkowicz, 2013; Xie and Zhang, 2017). In the classical way, the derivation of the wave equation in the PML and that of the boundary and/or interface conditions are independent, and the two may be improperly matched, which can lead to numerical instability and reduced numerical accuracy, as shown by Duru and Kreiss (2014). However, in taking the weak-form approach, such a mismatch can be naturally avoided because the free and interface conditions are simultaneously stretched in a consistent way. In the following sections, we derive the second-order M-UFSPM using the weak-form approach. We start from the second-order wave equation in the weak form specified in the infinite domain $\{x>0, y>0\}$, which is written as:

$$\begin{aligned}&\int_\Omega \rho\left(-\omega^2\right)\left(\hat{u}_x w_x+\hat{u}_y w_y\right)\mathrm{d}x\mathrm{d}y+\\&\int_\Omega\left(\partial_x w_x\hat{\sigma}_{xx}+\partial_y w_x\hat{\sigma}_{xy}+\partial_x w_y\hat{\sigma}_{yx}+\partial_y w_y\hat{\sigma}_{yy}\right)\mathrm{d}x\mathrm{d}y\\&=\oint_\Gamma\left[\left(w_x\hat{\sigma}_{xx}+w_y\hat{\sigma}_{yx}\right)\mathrm{d}y+\left(w_x\hat{\sigma}_{xy}+w_y\hat{\sigma}_{yy}\right)\mathrm{d}x\right]\end{aligned} \tag{13}$$

where Ω denotes the whole domain in $\{x>0, y>0\}$ and Γ represents the boundary of Ω. We map Eq. (13) into complex coordinates defined by:

$$\begin{aligned}\tilde{x}(x,y)&=x+\int_0^x\left\{\left[d(x)+p^{x/y}d(y)\right]/\left[\alpha(x)+\mathrm{i}\omega\right]\right\}\mathrm{d}x'\\&=x+\int_0^x\left\{d_x^m/\left[\alpha_x+\mathrm{i}\omega\right]\right\}\mathrm{d}x'\end{aligned} \tag{14}$$

$$\begin{aligned}\tilde{y}(x,y)&=y+\int_0^y\left\{\left[p^{y/x}d(x)+d(y)\right]/\left[\alpha(y)+\mathrm{i}\omega\right]\right\}\mathrm{d}y'\\&=y+\int_0^y\left\{d_y^m/\left[\alpha_y+\mathrm{i}\omega\right]\right\}\mathrm{d}y'\end{aligned} \tag{15}$$

and transformed into real coordinates using the incorrect chain rule Eq. (3), thereby obtaining:

$$\begin{aligned}&\int_\Omega\rho\left(\tilde{\hat{u}}_x w_x+\tilde{\hat{u}}_y w_y\right)\mathrm{d}x\mathrm{d}y+\\&\int_\Omega\left(\partial_x w_x\tilde{\hat{\sigma}}_{xx}+\partial_y w_x\tilde{\hat{\sigma}}_{xy}+\partial_x w_y\tilde{\hat{\sigma}}_{yx}+\partial_y w_y\tilde{\hat{\sigma}}_{yy}\right)\mathrm{d}x\mathrm{d}y\\&=\oint_\Gamma\left[\left(w_x\tilde{\hat{\sigma}}_{xx}+w_y\tilde{\hat{\sigma}}_{yx}\right)\mathrm{d}y+\left(w_x\tilde{\hat{\sigma}}_{xy}+w_y\tilde{\hat{\sigma}}_{yy}\right)\mathrm{d}x\right]\end{aligned} \tag{16}$$

$$\tilde{\hat{u}}_x=\left(-\omega^2\right)s_x s_y\hat{u}_x,\ \tilde{\hat{u}}_y=\left(-\omega^2\right)s_x s_y\hat{u}_y \tag{17}$$

$$\begin{aligned}\tilde{\hat{\sigma}}_{xx}&=c_{11}\left(s_y/s_x\right)\partial_x\hat{u}_x+c_{12}\partial_y\hat{u}_y,\\\tilde{\hat{\sigma}}_{xy}&=c_{66}\left(s_x/s_y\right)\partial_y\hat{u}_x+c_{66}\partial_x\hat{u}_y\end{aligned} \tag{18}$$

$$\begin{aligned}\tilde{\hat{\sigma}}_{yx}&=c_{66}\partial_y\hat{u}_x+c_{66}\left(s_y/s_x\right)\partial_x\hat{u}_y,\\\tilde{\hat{\sigma}}_{yy}&=c_{12}\partial_x\hat{u}_x+c_{22}\left(s_x/s_y\right)\partial_y\hat{u}_y\end{aligned} \tag{19}$$

where $s_x=1+d_x^m/\left(\alpha_x+\mathrm{i}\omega\right)$ and $s_y=1+d_y^m/\left(\alpha_y+\mathrm{i}\omega\right)$. The inverse Fourier transformations of $\left(-\omega^2\right)s_x s_y$, s_x/s_y, and s_x/s_y can be formally written as:

$$\begin{aligned}\mathcal{F}^{-1}\left[\left(-\omega^2\right)s_x s_y\right]&=\ddot{\delta}(t)+a_0\dot{\delta}(t)+\\&a_1\delta(t)+a_2\mathrm{e}^{-\alpha_x t}H(t)+a_3\mathrm{e}^{-\alpha_y t}H(t)\end{aligned} \tag{20}$$

$$\mathcal{F}^{-1}\left[s_x/s_y\right]=\delta(t)+a_4\mathrm{e}^{-\alpha_x t}H(t)+a_5\mathrm{e}^{-\left(\alpha_y+d_y^m\right)t}H(t) \tag{21}$$

$$\mathcal{F}^{-1}\left[s_y/s_x\right]=\delta(t)+a_6\mathrm{e}^{-\alpha_y t}H(t)+a_7\mathrm{e}^{-\left(\alpha_x+d_x^m\right)t}H(t) \tag{22}$$

The detailed expression of a_i, $i=1,\cdots,7$ can be found in Appendix B of the study of Xie *et al.* (2014). Substituting Eqs. (20)–(22) with Eqs. (17)–(19), we obtain the time-domain M-UFSPML in its weak form:

$$\begin{aligned}&\int_\Omega\rho\left(\tilde{\tilde{u}}_x w_x+\tilde{\tilde{u}}_y w_y\right)\mathrm{d}x\mathrm{d}y+\\&\int_\Omega\left(\partial_x w_x\tilde{\tilde{\sigma}}_{xx}+\partial_y w_x\tilde{\tilde{\sigma}}_{xy}+\partial_x w_y\tilde{\tilde{\sigma}}_{yx}+\partial_y w_y\tilde{\tilde{\sigma}}_{yy}\right)\mathrm{d}x\mathrm{d}y\\&=\oint_\Gamma\left[\left(w_x\tilde{\tilde{\sigma}}_{xx}+w_y\tilde{\tilde{\sigma}}_{yx}\right)\mathrm{d}y+\left(w_x\tilde{\tilde{\sigma}}_{xy}+w_y\tilde{\tilde{\sigma}}_{yy}\right)\mathrm{d}x\right]\end{aligned} \tag{23}$$

Taking $\tilde{\tilde{u}}_x$ as an example, we find

$$\begin{aligned}\tilde{\tilde{u}}_x&=\left(-\omega^2\right)s_x s_y\tilde{\tilde{u}}_x=\ddot{u}_x+a_0\dot{u}_x+a_1u_x+\\&a_2\left[\mathrm{e}^{-\alpha_x t}H(t)\right]\cdot u_x+a_3\left[\mathrm{e}^{-\alpha_y t}H(t)\right]\cdot u_x\end{aligned} \tag{24}$$

within which two convolution terms exist. To compute convolution terms, we need to introduce a memory

variable such as $\psi(t)=\left[\mathrm{e}^{-bt}H(t)\right]\cdot g(t)$. Denoting the evaluation of $\psi(t)$ at $t_n=n\Delta t$ by ψ_n, ψ_{n+1} can be updated using the recursive convolution technique summarized by Xie *et al*. (2014). Thus, 12 memory variables are needed to deal with all the convolution terms in the time-domain M-UFSPML. To reduce the number of memory variables, we restructure the M-UFSPML as:

$$\begin{aligned}&\int_\Omega \rho\left(\tilde{\tilde{u}}_x w_x+\tilde{\tilde{u}}_y w_y\right)\mathrm{d}x\mathrm{d}y+\\&\int_\Omega\left(\partial_x w_x\tilde{\tilde{\sigma}}_{xx}+\partial_y w_x\tilde{\tilde{\sigma}}_{xy}+\partial_x w_y\tilde{\tilde{\sigma}}_{yx}+\partial_y w_y\tilde{\tilde{\sigma}}_{yy}\right)\mathrm{d}x\mathrm{d}y\\&=\oint_\Gamma\left[\left(w_x\tilde{\tilde{\sigma}}_{xx}+w_y\tilde{\tilde{\sigma}}_{yx}\right)\mathrm{d}y+\left(w_x\tilde{\tilde{\sigma}}_{xy}+w_y\tilde{\tilde{\sigma}}_{yy}\right)\mathrm{d}x\right]\end{aligned}\tag{25}$$

$$\begin{aligned}\tilde{\tilde{u}}_x&=\ddot{u}_x+a_0\dot{u}_x+a_1u_x+a_2\overline{u}_x^{\alpha_x}+a_3\overline{u}_x^{\alpha_y},\\\tilde{\tilde{u}}_y&=\ddot{u}_y+a_0\dot{u}_y+a_1u_y+a_2\overline{u}_y^{\alpha_x}+a_3\overline{u}_y^{\alpha_y}\end{aligned}\tag{26}$$

$$\tilde{\tilde{\sigma}}_{ij}=\sigma_{ij}-d_j^m\overline{\sigma}_{ij}^{d_j^m+\alpha_j},\ i,j=x,y\tag{27}$$

$$\begin{aligned}\sigma_{xx}&=c_{11}\partial_x\left(u_x+d_y^m\overline{u}_x^{\alpha_y}\right)+c_{12}\partial_y\left(u_y+d_x^m\overline{u}_y^{\alpha_x}\right),\\\sigma_{xy}&=c_{66}\partial_y\left(u_x+d_x^m\overline{u}_x^{\alpha_x}\right)+c_{66}\partial_x\left(u_y+d_y^m\overline{u}_y^{\alpha_y}\right)\end{aligned}\tag{28}$$

$$\begin{aligned}\sigma_{yx}&=\sigma_{xy},\\\sigma_{yy}&=c_{12}\partial_x\left(u_x+d_y^m\overline{u}_x^{\alpha_y}\right)+c_{22}\partial_y\left(u_y+d_x^m\overline{u}_y^{\alpha_x}\right)\end{aligned}\tag{29}$$

where $\overline{u}_i^{\alpha_j}=\left[\mathrm{e}^{-\alpha_j t}H(t)\right]\cdot u_i(t)$,

$$\overline{\sigma}_{ij}^{d_j^m+\alpha_j}=\left[\mathrm{e}^{-\left(d_j^m+\alpha_j\right)t}H(t)\right]\cdot\sigma_{ij}(t),\ i,j=x,y$$

Through simple counting, we see that only eight memory variables are needed in the restructured M-UFSPML. In the three-dimensional case, only 18 memory variables are needed in the restructured M-UFSPML, whereas in the original formulation, 39 memory variables were needed. However, in the case that $\alpha_i=0$, the number of additionally required memory variables in the M-UFSPML, as presented in Eqs. (25)–(29) is greater than that in the M-UFSPML presented in Eqs. (16)–(19), which comprises four in the two-dimensional case.

For comparison, we summarize the M-PML formulations and the number of additional variables needed for their implementation in time-domain elastodynamics, as shown in Table 1. The number listed in the table is the number of additional variables introduced in the M-PML relative to the wave equation in the computational domain. Notably, in the first-order case, there are five wave equations (two for velocities and three for stresses), whereas in the second-order case there are two wave equations (two for displacements). The table thus shows that the memory cost is lowest for our formulation. Moreover, in their M-USFPML, Li and Bou (2010) treated the convolutional terms involving the divergence of stress and containing second-order space partial differentials as body force terms. The direct computation of second-order space partial differentials in the FE or SE method requires a dense grid in a case in which high-order (≥ 2) element interpolant functions are used, and such computation could be wrong in the case of low-order (≤ 2) element interpolant functions (Zhebel *et al*., 2014). The same problem exists for the first-order M-UFSPML given by Martin *et al*. (2008). Thus, the M-UFSPML presented here is more suitable for implementation using FE/SE methods, irrespective of the order of element interpolants. The weak-form M-UFSPML shares the same structure as the weak-form wave equation, and the implementation of the M-UFSPML can thus use the same code as the wave equation, except for the additional treatment of convolution within the element-wise stress and strain computation. A detailed implementation of the second-order wave equation was described by Komatitsch and Tromp (1999), and we do not repeat it in this paper.

4 Numerical experiments

We present three numerical experiments to test the accuracy and long-term stability of our M-UFSPML formulation. The first example involves isotropic wave simulation in a homogeneous semi-infinite model. The second involves anisotropic wave simulations in a homogeneous infinite model. The third involves simulation using a horizontally layered model filled with coupled isotropic and anisotropic media. We compare the accuracy of our M-UFSPML against that of the

Table 1 M-PML formulations and a number of additional variables necessary for their implementation in time-domain elastodynamics (in-plane wave)

	First order	Second order
Split	Meza-Fajardo and Papageorgiou (2008)#; 10#	Ping *et al*. (2014, 2016)#; 10#
	Gao and Huang (2017)#; 10	
Unsplit	Martin *et al*. (2008)*; 8*	Li and Bou (2010)#*; 8#(8*)
	Gao and Huang (2017)*; 8*	Fathi *et al*. (2015)#; 6#
		This paper#*; 4#(8*)

(# or * denotes the use of a frequency-shifted or classical coordinate stretching function)

extended-domain solution obtained by enlarging the truncated domain, in addition to other results obtained using different infinite-domain truncation techniques. A fixed-boundary condition (setting the displacement to zero) is applied along the outside boundary of the M-UFSPML.

Unless otherwise specified, in all experiments we use a polynomial degree $N = 5$ for the Lagrange interpolants within each Legendre spectral element. For time-stepping, we use the second-order explicit Newmark scheme ($\beta = 0$, $\gamma = 1/2$), combined with a recursive convolution technique as given in Eq. (73) of the paper published by Xie *et al*. (2014). According to Collino and Tsogka (2001), the damping profile that is normal for the M-UFSPML interface is

$$d(x_i) = \frac{3V_{P,\max}}{2L}\left(|x_i|/L\right)^2 \tag{30}$$

where L is the thickness of the M-UFSPML, x_i is the distance along the normal direction measured from the interface, and $V_{P,\max}$ is the maximum velocity of the primary arriving wave in the media. Following Zhang and Shen (2010), we set

$$\alpha(x_i) = 3\pi f_0\left(1 - |x_i|/L\right) \tag{31}$$

where f_0 is the dominant frequency of the source time function,

$$R(t) = \left[1 - 2\left(\pi f_0\left(t - t_0\right)\right)^2\right] \mathrm{e}^{\left[-\left(\pi f_0\left(t-t_0\right)\right)^2\right]} \tag{32}$$

and t_0 is its onset time. We study the decay of energy with time throughout the domain to check the long-term stability of our formulation by evaluating total energy, which is the sum of the kinetic and potential energies:

$$E = \frac{1}{2}\rho\left\|\dot{\boldsymbol{U}}\right\|_2 + \frac{1}{2}\sum_e\sum_j \sigma_{lk}^{e,j}\varepsilon_{lk}^{e,j} \tag{33}$$

where $\left\|\dot{\boldsymbol{U}}\right\|_2$ is the norm of the velocity vector and $\sigma_{lk}^{e,j}$, $\varepsilon_{lk}^{e,j}$ are respectively the stress and strain components.

4.1 Isotropic wave simulation in a semi-infinite domain

We consider a homogeneous isotropic semi-infinite

Table 2 Geometrical and discretization parameters for the homogeneous isotropic semi-infinite model and the anisotropic infinite model

	Isotropic	Anisotropic medium III, IV	Apatite
Physical domain dimension			
Length	11 km	8 m	25 cm
Width	4 km	8 m	25 cm
Physical domain properties			
Density	2700 kg/m^3	1 kg/m^3	3200 kg/m^3
Elastic coefficients	V_P =3200 m/s, V_S =1870 m/s.	Given by Table 3	Given by Table 3
Source location			
From left boundary	0.25 km	4 m	12.5 cm
From bottom boundary	3.75 km	4 m	12.5 cm
Receiver location			
From left boundary	9.8 km	5 m	22.5 cm
From bottom boundary	4 km	6 m	20.5 cm
Ricker wavelet parameters			
Dominant frequency	2.5 Hz	1 Hz	170 kHz
Onset time	0.4 s	0.5 s	5.88 μs
Discretization parameters			
Element size	0.1 km	0.25 m	0.25 cm
Layer of elements used in PML	3, 9, 15	9	9
Time step	0.0004 s	0.001 s	0.02 μs
Total duration	20 s	100 s	2000 μs

model with a traction-free boundary at its top. The M-UFSPML is applied for infinite-domain truncation along the other three sides. The elastic properties of the medium and geometrical and discretization parameters of the model are listed in the second column of Table 2. Within the M-UFSPML, we set $p^{x|y}$ and $p^{y|x}$ equal to 0.1. Owing to the loss of the perfect-matching property of the M-UFSPML, we first compare the accuracy of the normalized displacement of the USFPML, with a thickness of 300 m, with that of the M-UFSPML, which has different thicknesses, namely 300, 900 and 1500 m, corresponding to 3, 9 and 15 elements, respectively. For the P wave, the respective ratios of the M-UFSPML thickness to the dominant wavelength are approximately 0.23, 0.70 and 1.17. For the S wave, the ratios are approximately equal to 0.40, 1.20 and 2.00. Figure 3 shows snapshots of the wavefield at 2, 4, and 8 s, computed with the nine-element M-UFSPML and three-element UFSPML. Owing to the unified scale used for all snapshots, we do not see a clear difference between the snapshots obtained with the nine-element M-UFSPML and the three-element UFSPML. However, as seen in Fig. 4, we see the time history of the normalized displacement at the receiver, revealing that the accuracy of the M-UFSPML increases with the thickness of the M-UFSPML and is always lower than that of the UFSPML. Only in the case of a thickness of 15 elements is the accuracy of the M-UFSPML close to that of the UFSPML. The accuracy of the nine-element M-UFSPML is sufficiently fine for practical application.

The M-UFSPML has no clear difference in its absorbing efficiency between body waves and Rayleigh waves. We compare the accuracy of the M-UFSPML in our paper with that of the M-UFSPML obtained by Li and Bou (2010), both having a thickness set of nine elements. Figure 5, comparing the time history of the displacement at the receiver, shows that the accuracy of the M-UFSPML is slightly better than that of the M-UFSPML obtained by Li and Bou (2010), referred to as the M-UFSPML-Li. Additionally, the accuracy of the M-UFSPML is better than that of the M-UPML, within which $\alpha(x_i)=0$ and the widely used first-order Stacey absorbing boundary condition is present (Stacey, 1988). To ensure a fair comparison, in the event that the Stacey absorbing boundary condition is used, the computation domain is extended to include the space occupied by the M-UFSPML.

4.2 Anisotropic wave simulation in infinite domain

We now consider the anisotropic wave simulation in the homogeneous infinite domain. Three types of anisotropic medium are considered. The geometry and discretization parameters are given in the second and third columns of Table 2. The elastic properties of the anisotropic medium are given in Table 3. In all cases, we set $p^{x|y}$ and $p^{y|x}$ of the M-UFSPML equal to 0.1, except for anisotropic medium III, for which we use $p^{x|y}=p^{y|x}=0.25$. For each case, we compare the absorbing efficiency of the M-UFSPML, of which the

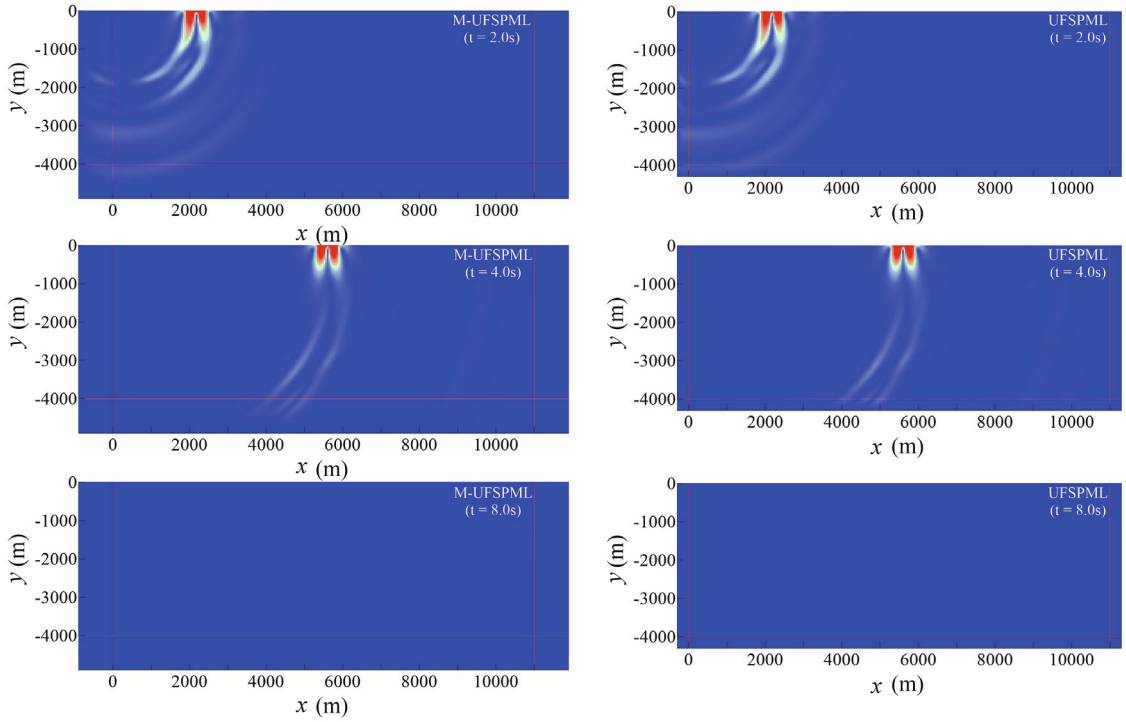


Fig. 3 Snapshots of the normalized displacement $\sqrt{u_x^2+u_y^2}$ taken at 2, 4, and 8 s, computed with different infinite-domain truncation techniques: (left column) M-UFSPML with a thickness of nine elements and (right column) UFSPML with a thickness of three elements (isotropic medium, semi-infinite model, vertical excitation)

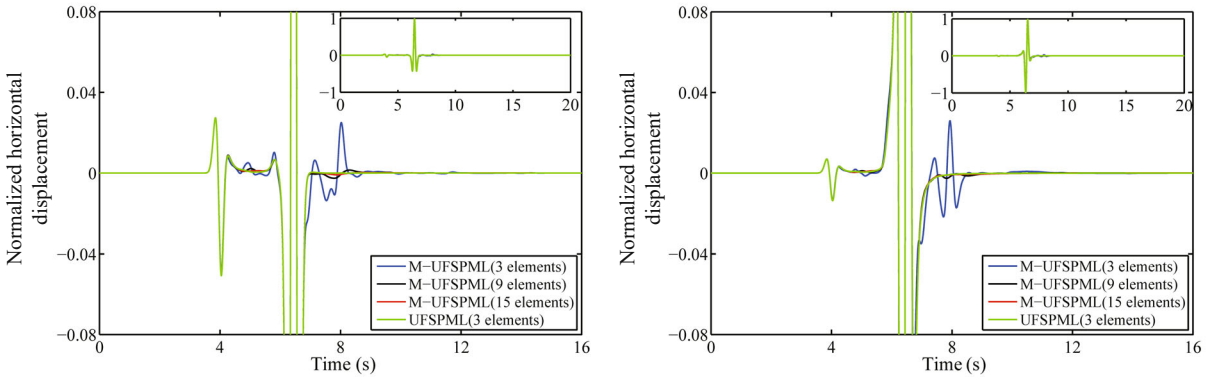


Fig. 4 Comparison of time histories of the normalized horizontal and vertical displacements at the receiver, computed with M-UFSPMLs having thicknesses of 3, 9, and 15 elements and an UFSPML having a thickness of three elements (isotropic medium, semi-infinite model, vertical excitation)

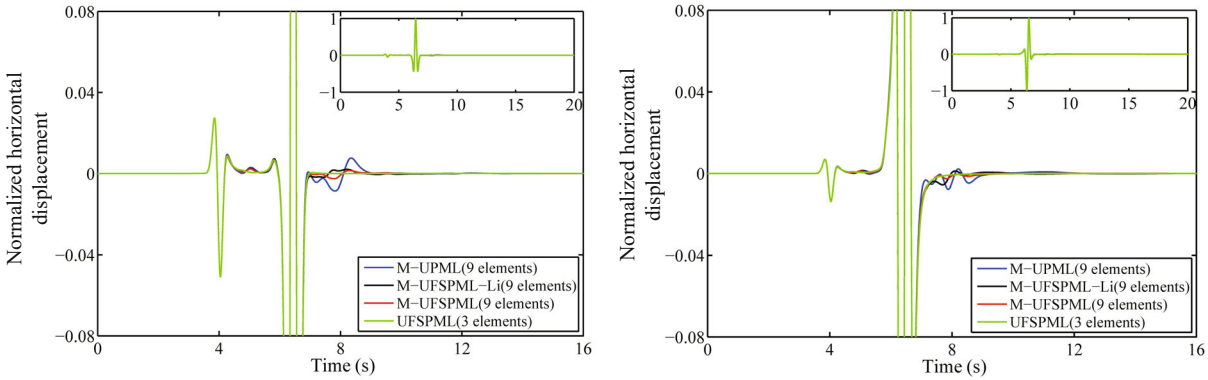


Fig. 5 Comparison of time histories of the normalized horizontal and vertical displacements at the receiver, computed with an M-UFSPML, an M-UPML, and an M-UFSPML-Li having a thickness of nine elements and an UFSPML having a thickness of three elements. M-UFSPML-Li refers to the result obtained with the M-UFSPML used by Li and Bou (2010) (isotropic medium, semi-infinite model, vertical excitation)

thickness is set uniformly as nine times the element size, with the efficiency of the standard Stacey ABC. The wave speed in the anisotropic medium depends on the azimuth, and the ratios of the M-UFSPML thickness to the dominate wavelength are thus different for waves having a different azimuth—that is, approximately 0.7 for primary arriving waves and 1.2 for secondary arriving waves. To ensure a fair comparison, in the case of the Stacey ABC, the computation domain is extended to include the space occupied by the M-UFSPML. In all cases, we first present snapshots of the normalized displacement at three different times for a vertical excitation. The left panel of Fig. 7 shows waves at an evolving stage, whereas the center and right panels show the absorption of waves in the M-UFSPML region. In Figs. 7, 9, and 11 there are no discernible reflections from the M-UFSPML interface, indicating satisfactory performance of the nine-element M-UFSPML. Figures 10, 12, and 14 compare the normalized horizontal and vertical displacement time histories at the receiver, computed with the M-UFSPML, Stacey ABC, and infinitely extended domain, the solution for which is referred to as an extended solution. We observe that relatively good absorbing efficiency is achieved with the nine-element M-UFSPML, which is less than 2% in the case of medium III and less than 1% in other cases. The excellent performance of the M-UFSPML is partially attributable to the non-grazing incident waves in all cases.

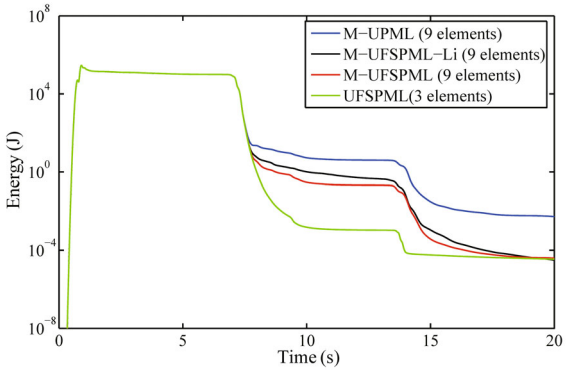


Fig. 6 Total decay of energy throughout the domain, computed with the M-UFSPML, M-UPML, and M-UFSPML-Li, having a thickness of nine elements and the UFSPML, having a thickness of three elements. M-UFSPML-Li refers to the result obtained by using the M-UFSPML by Li and Bou (2010) (isotropic medium, semi-infinite model, vertical excitation)

Table 3 Elasticity coefficients of an anisotropic medium

Elasticity coefficients (N/m^2)	III	IV	Apatite
c_{11}	4	10	1.65×10^{11}
c_{22}	20	20	6.20×10^{10}
c_{66}	2	6	3.96×10^{10}
c_{12}	7.5	2	5.00×10^{10}

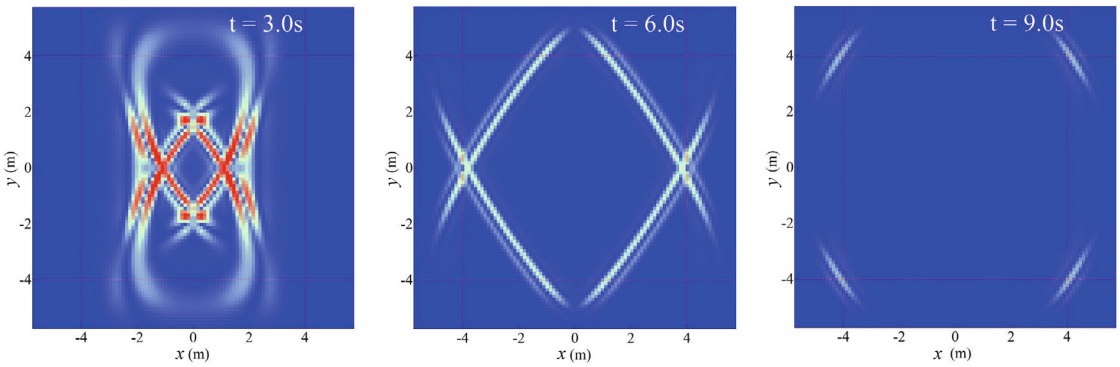


Fig. 7 Comparison of snapshots of the normalized displacement $\sqrt{u_x^2+u_y^2}$ taken at 3, 6, 9 s and computed with different infinite-domain truncation techniques: M-UFSPML having a thickness of nine elements (anisotropic medium III, infinite model, vertical excitation)

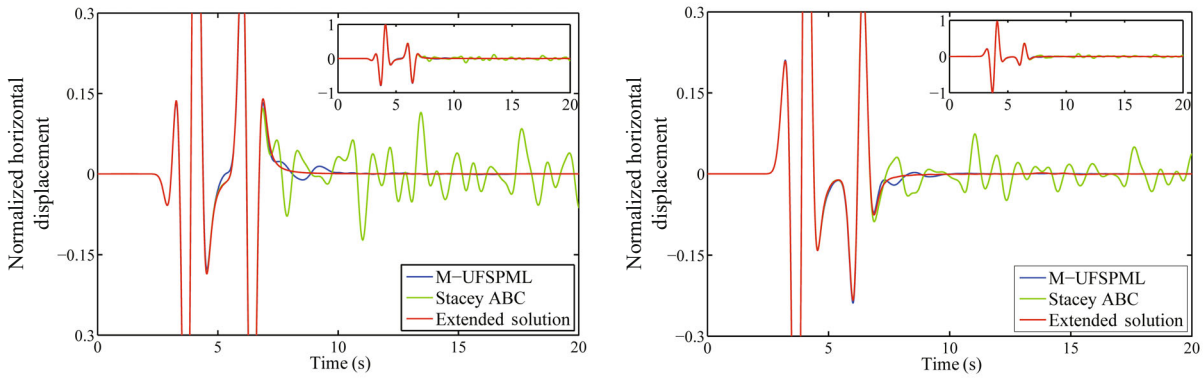


Fig. 8 Comparison of time histories of the normalized horizontal and vertical displacements at the receiver, computed with the M-UFSPML having a thickness of nine elements, Stacey ABC and the extended domain (anisotropic medium III, infinite model, vertical excitation)

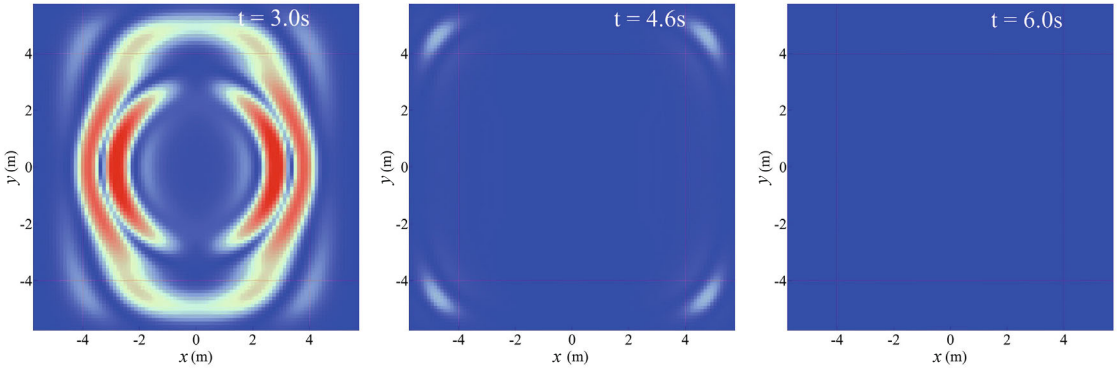


Fig. 9 Comparison of snapshots of the normalized displacement $\sqrt{u_x^2+u_y^2}$ taken at 3.0, 4.6, and 6.0 s and computed with different infinite-domain truncation techniques: M-UFSPML having a thickness of nine elements (anisotropic medium IV, infinite model, vertical excitation)

Reflection can also be observed owing to the imperfect interface of the M-UFSPML. We therefore recommend using the UFSPML for an anisotropic medium with elastic properties, satisfying the sufficient conditions for PML stability given by Bécache *et al*. (2003).

4.3 Coupled isotropic and anisotropic wave simulation in a layered model

We present the performance of the M-UFSPML in a heterogeneous medium. We consider a two-dimensional, four-layer model adapted from the work of Zhu (2017), with the medium properties given in Table 4. The media in the first and fourth layers are isotropic. In the second layer, a medium having vertical transverse isotropy is considered. To consider the main anisotropic medium used in seismology, the medium in the third layer is set as a single set of vertical fractures embedded in an isotropic medium, resulting in horizontal transverse isotropy. The discretization parameters are given in

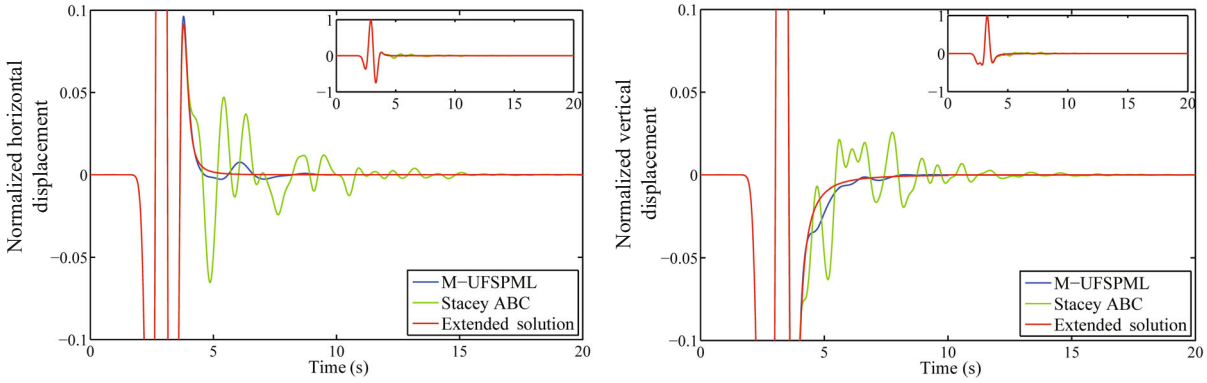


Fig. 10 Comparison of time histories of normalized horizontal and vertical displacements at the receiver, computed with M-UFSPML having a thickness of nine elements, Stacey ABC and the extended domain (anisotropic medium IV, infinite model, vertical excitation)

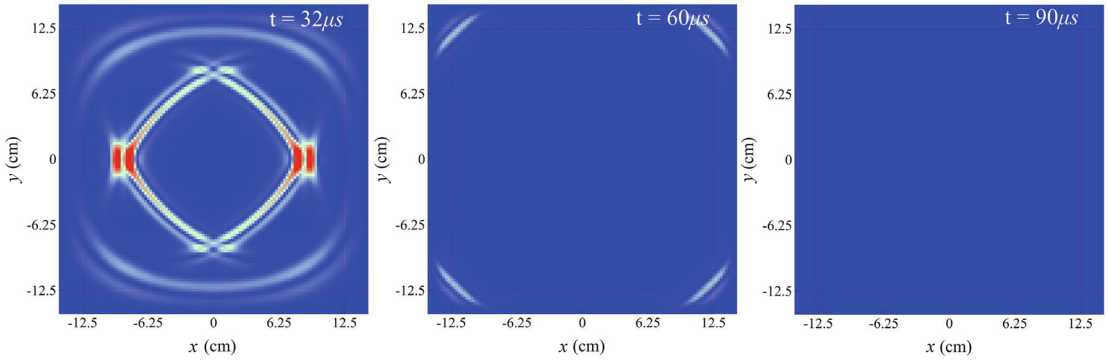


Fig. 11 Comparison of snapshots of the normalized displacement $\sqrt{u_x^2 + u_y^2}$ taken at 32, 60, and 90 μs and computed with different infinite-domain truncation techniques: M-UFSPML having a thickness of nine elements (anisotropic medium Apatite, infinite model, vertical excitation)

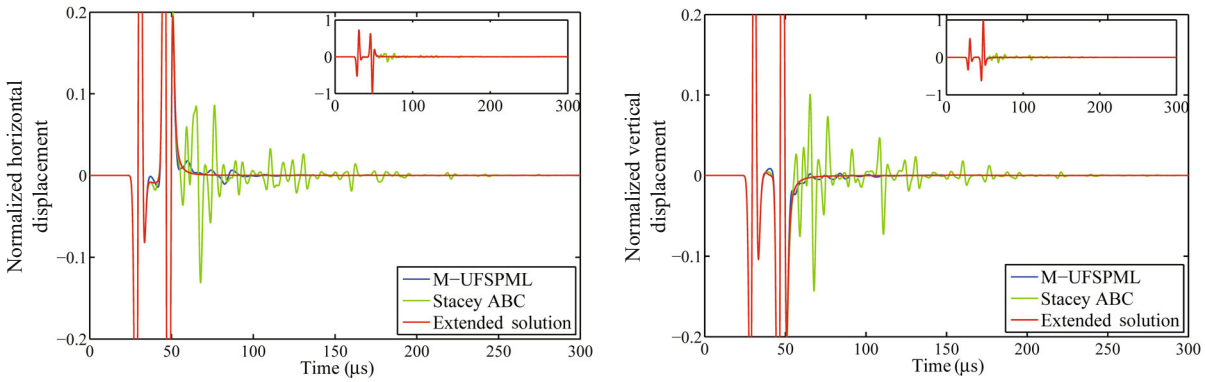


Fig. 12 Comparison of normalized horizontal and vertical displacement time histories at the receiver, computed with M-UFSPML, having a thickness size of 9 elements, Stacey ABC and the extended domain (anisotropic medium apatite, infinite model, vertical excitation)

Table 5. We set $p^{x|y}$ and $p^{y|x}$ of the M-UFSPML at 0.1 and the M-UFSPML thickness as nine elements. The wave speed in an anisotropic medium depends on the azimuth, and the ratios of the M-UFSPML thickness to the dominate wavelength thus differ between waves with different azimuth—that is, approximately 0.7 for primary arriving waves and 1.2 for secondary arriving waves. As shown in Fig. 13, the M-UFSPML performs well in absorbing complex waves in the layered model. A long-time numerically stable simulation can be conducted. Figure 14 shows the energy decay curve for the entire domain, containing the truncated computational domain and M-UFSPML. Additionally, we compare the energy decay curve, computed with Stacey ABC and the M-UFSPML given by Li and Bou (2010), in which the thickness is set to nine elements. The energy decay is more rapid when our M-UFSPML is used, owing to its better absorbing efficiency. However, from the time history of the two components of the normalized displacement at the receiver 100 m from the right of the M-UFSPML, as shown in Fig. 15, clear spurious reflections returning to the physical domain could be observed in all cases owing to near-grazing waves generated by multiple reflections and transmission. Finally, it is worth noting that the long penetration depth associated with surface waves must be considered to avoid their interference with the M-UFSPML. The same problem exists for PML application. Xie *et al*. (2016) illustrated that for PML application in wave simulation within an elastic Pekeris waveguide, when surface waves interfere with the PML,

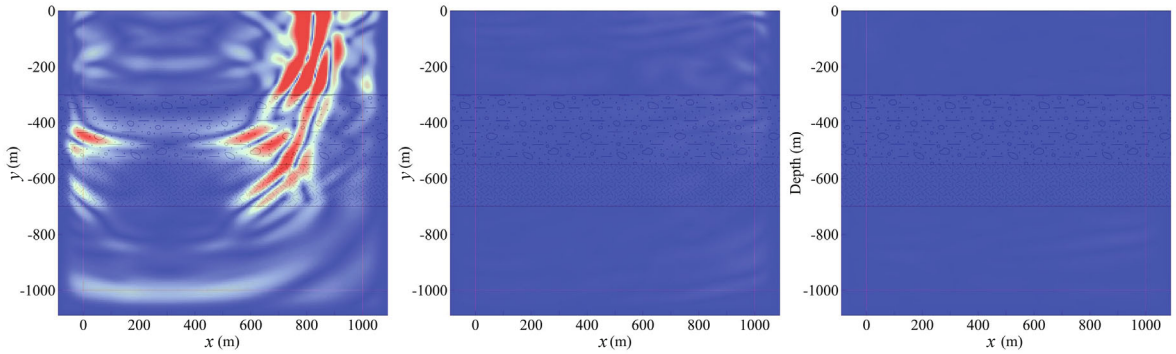


Fig. 13 Comparison of snapshots of the normalized displacement $\sqrt{u_x^2+u_y^2}$ taken at 0.6 s (left), 1.2 s (center), and 1.8 s (right), computed with different infinite-domain truncation techniques: M-UFSPML having a thickness of nine elements (coupled isotropic and anisotropic medium, layered model, vertical excitation)

Table 4 Media properties of the adapted layered model, taken from Zhu (2017)

Layers	Thickness	V_P (m/s)	V_S (m/s)	ρ (kg/m^3)	c_{11} (N/m^2)	c_{22} (N/m^2)	c_{66} (N/m^2)	c_{12} (N/m^2)
1	300 m	1900	1200	1800	–	–	–	–
2	250 m	–	–	2300	2.33×10^{10}	1.32×10^{10}	2.78×10^{9}	6.56×10^{9}
3	150 m	–	–	2200	1.35×10^{10}	1.98×10^{10}	8.8×10^{9}	1.04×10^{10}
4	–	3200	1500	2500	–	–	–	–

Table 5 Geometrical and discretization parameters for the adapted layered model, taken from Zhu (2017)

Truncated computational domain dimension		Receiver location		Discretization parameters	
Length	1000 m	From left boundary	900 m	Polynomial degree of SE	5
Width	1000 m	From bottom boundary	100 m	Element size	10 m
Source location		Ricker wavelet parameters		Layer of elements used in PML	9
From left boundary	300 m	Dominant frequency	10 Hz	Timestep	0.0002 s
From bottom boundary	900 m	Onset time	0	Total duration	20 s

the absorbing efficiency of the PML decreases and cannot be improved by simply increasing the thickness of the PML.

5 Conclusions

We developed a second-order M-UFSPML for infinite-domain truncation in the simulation of a linear anisotropic wave. Using plane wave analysis, we validate the importance of the usage of the non-rigorous chain rules in Meza-Fajardo and Papageorgiou (2008), which had not previously been addressed. The non-rigorous chain rules, rather than the rigorous ones, improve the absorbing efficiency of M-PML for low-frequency incident waves, which is important for low-frequency seismic wave simulation adopted in seismic hazard analysis. Moreover, we show the importance of the complex frequency-shifted stretching function also can improve the absorbing efficiency of M-PML for near-grazing incident waves. Based on the above inspection, our M-UFSPML is derived through the multi-axial complex stretching of the second-order anisotropic wave equation written in its weak form. The formulation readily works with standard displacement-based FE and SE wave simulation, with only minor modification of the code in the computational domain. It is simple to implement and allows for a natural application of traction-free boundary or interface conditions. The accuracy and long-time stability of the formulation was demonstrated using high-order SE discretization with a second-order explicit Newmark time integrator, which showed the accuracy of the method and its ability to solve wave propagation in anisotropic media having orthorhombic symmetry. Although the interface of the M-PML is not perfectly matched, it remains more efficient at absorbing both body waves and interface waves than do low-order absorbing boundary conditions, such as the widely used Stacey ABC in the field of engineering. One topic of interest for future study would be to extend this work to forward and adjoint wave simulation in anisotropic viscoelastic media (Carcione, 2015). Another topic of

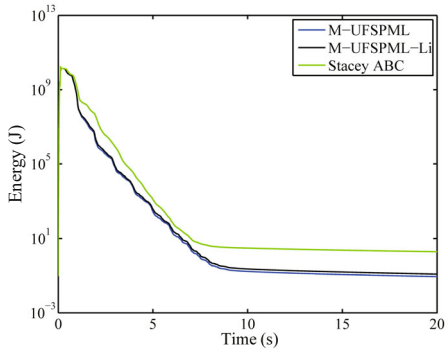


Fig. 14 Total decay of energy throughout the domain, computed with the M-UFSPML, M-UFSPML-Li, having a thickness of nine elements and Stacy ABC (coupled isotropic and anisotropic medium, layered model, vertical excitation). M-UFSPML-Li refers to the result obtained with the M-UFSPML by Li and Bou (2010)

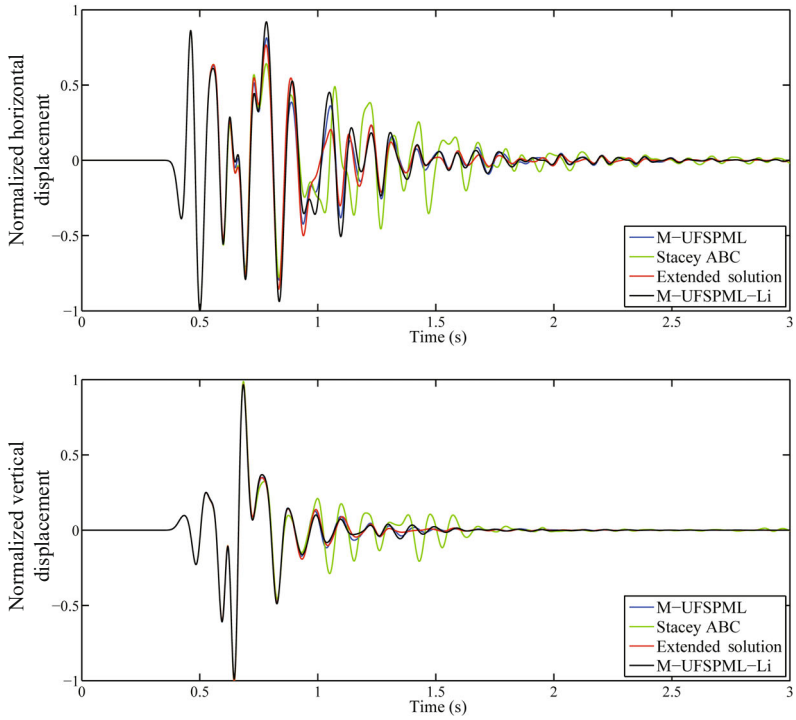


Fig. 15 Comparison of time histories of normalized horizontal and vertical displacements at the receiver, computed with the M-UFSPML, M-UFSPML-Li with a thickness of nine elements, Stacey ABC and the extended solution (coupled isotropic and anisotropic medium, layered model, vertical excitation). M-UFSPML-Li refers to the result obtained with the M-UFSPM by Li and Bou (2010)

interest for future studies would be to set up a community model for an accurate comparison of the computational efficiency of different infinite-domain truncation technologies (e.g., Liao *et al*., 1984; Panji *et al*., 2014; Zhao *et al*., 2018).

Acknowledgement

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